

Anruddh Pratap Upadhyay

Senior RF Test & NPI Process Engineer | Consumer Electronics EMS Expert

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Professional Summary

Strategic RF Test & NPI Process Engineer with 6+ years of comprehensive experience across R&D, EMS, and high-volume smartphone manufacturing environments (Apple, Vivo, Oppo, DBG). Specialised in RF test fixture design, hardware validation, advanced RF measurements, Camera, audio & MMI Test SOP and defects analysis to take preventive actions, and NPI-to-MP process optimisation. Proven expertise in cross-functional collaboration, rapid debugging, and delivering high-reliability test solutions for global smartphone platforms. Recognised for driving process improvements, yield enhancement, and technical leadership in fast-paced NPI environments.

Professional Experience

Apple R&D India (via LTTS) | Bengaluru, India

Sr. NPI RF TE/Process Engineer | Feb 2025 – Present

- RF Test Fixture Design & Validation: Designed and validated RF test fixtures for iPhone NPI builds (Cellular, WiFi, BT, RFID, Wireless Charging)
- Design Documentation: Delivered complete fixture wiring diagrams, signal routing schematics, and conducted design reviews
- PCB/MLB Design & review: PCB/MLB circuit's RF test points review and approval, Vendor engagement.
- RF Shield box IQC : RF Shield box IQC test such as isolation for qualification
- DFM : Strain gauge test of RF fixtures with MLB/PCB to avoid any physical damage during test.
- Cross-functional Support: Led CM debugging activities and failure analysis of test stations; supported global RF audit monitoring
- NPI Readiness: Ensured robust RF performance compliance across P1 → P2 → DVT/PVT/MP builds Designed and validated RF test fixtures for iPhone NPI builds (P0→P2→DVT/PVT/MP), covering Cellular, WiFi, BT, RFID, and Wireless Charging technologies.

Vivo Mobile India Pvt Ltd | Greater Noida, India

Senior NPI Electronics Process Engineer | Apr 2021 – Feb 2025

- NPI Leadership: Owned pilot builds from planning to first launch for new smartphone platforms
- RF Test Expertise: Specialised in FATP RF tester debugging, modem log analysis (QPST/AF Tool), and functional defect resolution
- Technical Validation: Conducted Network Spectrum Analysis using CMW and VNA for NFC/ antenna frame designs.
- Process Optimisation: Drove root cause analysis for PCB/LCM/CCM defects; validated local component trials
- Quality Improvements: Achieved 90% reduction in retest rate (20% → 2%) and increased pass rate to 99.63%

DBG Technology | Bawal, India

**Engineer-1* | Jul 2020 – Apr 2021*

- Assembly Line Management: Ensured uninterrupted production flow while maintaining quality standards
- Quality Systems: Implemented FMEA, corrective/preventive actions, and quality gate controls
- Defect Reduction: Drove RCA to minimise defect rates across multiple projects
- Recognition: Awarded "Best PIE Award" (September 2020)

Oppo Mobile India Pvt Ltd | Greater Noida, India

Graduate Engineer Trainee | Jun 2019 – Jul 2020

- Process Compliance: Executed machine/tool fool-proofing and MES scanning compliance
- Quality Assurance: Assisted QA in reducing market complaints through rigorous process audits

Core Competencies

RF Testing & Measurement

- Fixture Design: RF test fixture design, wiring diagrams, signal routing, shielding validation
- Conducted Testing: Antenna conducted tests, LTE/GSM/5G manual calling, S-parameter measurement
- Instrumentation: CMW500, Network Analyser (VNA), Spectrum Analyser, Signal Generator (STG)
- Wireless Technologies: Cellular (LTE/GSM), WiFi, BT, NFC, RFID, Wireless Charging
- OTA Measurements: TRP/TIS dark-room testing (hands-on China internship experience)

Hardware & Process Engineering

- Hardware Debug: PCB circuit analysis, component-level FA, X-ray inspection, LCR/CRO measurements
- NPI Process: Pilot build execution, MP readiness, GRR/MSA, Cpk/Cpp improvement, station qualification
- Quality Tools: 4M/5Y RCA, FMEA, statistical process control (JMP, Tableau)
- Test Development: Test scheme preparation, SWI documentation, fixture/jig design verification

Digital Interfaces: Working knowledge of SPI, I²C, and UART communication protocols for embedded and RF subsystem control, debugging, and validation.

Power Management (RF): Knowledge of DC-DC converters for RF applications, including buck/boost/LDO selection, noise mitigation, efficiency optimisation, and RF-sensitive PCB layout considerations.

Tools & Software

- Diagnostic: QPST, AF Tool, Serial Log Grab, AT Commands, Device/Modem log analysis
- Data Analysis: JMP, Tableau, MES systems
- Collaboration: OPP, BOX, RADAR, project management tools
- Kicad, Every-circuit PCB design Tool hands-on experience

Soft Skills

Cross-functional Leadership | Technical Communication | Rapid Problem Solving | Process Optimisation | Team Training & Mentoring

Education

B.Tech in Electronics & Communication Engineering

IMS Engineering College, AKTU University | 2019

Percentage: 72.5%

Class XII (Senior Secondary)

Dr. Ashok Kumar Smarak Inter College, Akbarpur | UP Board | 2014

Percentage: 87.8%

Class X (Secondary)

B L C P I C | UP Board | 2012

Percentage: 85.8%

Key Achievements & Certifications

- Six Sigma Black Belt – Lean & Six Sigma Fundamentals (Applied to NPI yield, FPY, and retest reduction)
- PMP® – Project Management Professional (Cross-functional NPI execution and delivery ownership)
- International Industrial Internship (Dongguan, China | 2024): Hands-on training in RF dark-room testing (TRP/TIS), fold display manufacturing, and advanced jig design processes.
- Vivo Fold Phone: Led RF and functional test design for company's first foldable smartphone platform.
- Process Improvements:
 - Improved functional pass rate by 1.13% (98.5% → 99.63%)
 - Reduced retest rate by 90% (20% → 2%)
 - Minimised man-made defects by 30% (1000 → 700 PPM)
- Awards: Best PIE Award (DBG Technology, 2020) | 3rd Prize in Techfest for IoT-based Smart Energy Meter (2018)
- Certification: WIPO Certification in Basic IPR

Languages

English (Professional Proficiency) | Hindi (Native)

I hereby declare that the information provided above is true to the best of my knowledge and belief.